Reg. No									

St. Joseph's College of Arts & Science (Autonomous)
St. Joseph's College Road, Cuddalore – 607001

EPPH1014 – MATERIAL SCIENCE

Time : 3 hrs **Max Marks :75**

SECTION – A (10X2=20)

Answer **ALL** Questions

1. What is bond energy?
2. Define Frank-Read Source.
3. What are solid solutions?
4. State the lever rule.
5. What is rate of transformation?
6. Mention any two applications of phase transformations.
7. What is energy gap?
8. What are the factors affecting electrical resistance of materials?
9. What is meant by resistivity?
10. Mention any two applications of dielectrics.

SECTION – B (5X5=25)

Answer **ALL** Questions

11. a) Write a brief note on classification of engineering materials.

(or)

b) What are point defects? Mention its types?

12. a) Explain in detail the Gibb's phase rule.

(or)

b) With a simple sketch explain the following systems
i) Eutecoid ii) Peritectic and iii) Peritectoid systems

13. a) Explain in detail the kinetics of homogeneous nucleation.

(or)

b) With a neat diagram explain the allotropy of iron.

14. a) Give an account on zone theory of solids.

(or)

b) Write a short note on zone in conductors, insulators and semi conductors.

15. a) Explain the classification of semiconductors on the basic of Fermi energy and Fermi levels.

(or)

b) What is piezoelectric effect? Explain the origin and mention few applications of piezoelectric crystals.

SECTION –C (3X10=30)

Answer any **THREE** Questions

16. Explain in detail the following bonding with examples and discuss the electrical properties.
    i) Ionic bond
    ii) Covalent bond
    iii) Metallic bond

17. Explain in detail the Hume-Rothery rule for the solid solubility of one element in another with few examples.

18. Explain in detail the transformation of Austenite upon continuous cooling by super imposing the cooling curves on TTT diagram.

19. Explain in detail the Drude and Lorentz theory of electrons.

20. What are Ferroelectric substances? Give the theory of ferroelectricity and mention its applications.

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